BuyerUnion

Describe what you need. Compare offers built around it.

Proposed product brief | Cal Hacks 13.0 | Visa: Reimagining Shopping | October 23, 2026

The product

Think of asking several stores for their best suitable offer without repeating your needs at every counter. BuyerUnion turns one shopping request into a small set of competing, checked offers. Each seller's AI finds relevant stock, explains its fit and may offer a discount within the seller's limits. The buyer sees the complete price, important tradeoffs and cited product evidence before approving anything.

Who it helps

Proposed first audience: students comparing laptops across participating electronics sellers. Today they must translate everyday needs into specifications, compare delivery and returns, and decide which advice to trust. Sellers must spend time finding suitable stock and deciding whether a concession is worthwhile.

The broader product covers manufactured consumer goods. Laptops are the primary demo; clothing, household products and appliances are extensions of the same request-and-offer flow. This brief proposes prioritizing one complete journey before expanding the live demo. That first audience and priority are recommendations, not validated demand or changes already approved by the team.

The experience

1. **Tell us what you need.** Use ordinary language. The assistant asks only questions that could change the offer.
2. **Confirm the brief.** Edit the requirements, delivery deadline and private maximum. Sharing a negotiation target is optional and requires consent.
3. **Receive offers.** Three isolated demo seller agents search their own catalogs and propose suitable products at permitted prices. A seller may hold its price or decline.
4. **Check what matters.** The platform checks required facts. One targeted challenge can correct a material claim; one final price round can improve an offer when useful.
5. **Compare the tradeoffs.** See up to three primary offers with all-in price, arrival, condition, a clear fit reason and one material concern. Expand for sources, service terms and price history.
6. **Approve the exact terms.** The selected offer is rechecked and temporarily reserved. Only the buyer can approve the final amount. The prototype ends with a clearly labeled simulated result unless a supported Visa test is verified.

Why this is different

The hypothesis is that buyers benefit when merchants respond to their needs instead of making them search every storefront. The distinctive demonstration combines seller-controlled concessions with checked requirements, independent product sources and buyer approval. AI negotiation already exists elsewhere; BuyerUnion must show that this combination improves an ordinary consumer decision.

What the hackathon demo must prove

One request, one understandable decision

Use one laptop request across three fictional merchants. Demonstrate a useful correction, then explain the resulting choice. For example, an agent must correct a product that misses a confirmed requirement rather than defend it with persuasive language. Use an explicit test case; show the actual outcome rather than guaranteeing a scripted correction.

The existing pricing example remains useful as a disclosed illustration:

Offer	Initial delivered total	Final delivered total	Meaning
Demo seller A, P3 laptop	$995	$975	$20 improvement; seller floor respected.
Demo seller B, same P3	$1,000	$970	$30 improvement; seller floor respected.
Demo seller C, P4 variant	$1,090	$1,090	No discount authorized; above this buyer's private ceiling.

The best available delivered price moves from $995 to $970: a $25 improvement within fictional demo offers. This is not proven retail savings. The buyer still chooses; price does not override required features or service differences. Illustrative prices are not guaranteed model outputs.

Keep the experience fast and clear

Proposed targets: first valid offers within 15 seconds and a useful final comparison within 45 seconds of confirming the brief. Measure cold and cached runs separately. These are usability targets, not achieved performance claims. Show partial results when research is slow and disclose any real cached evidence. Keep agent transcripts behind an optional detail view.

Each offer starts with the product and condition, all-in total, delivery, why it fits and its main concern. Supporting requirements, warranty/returns, cited evidence and initial-to-final price are expandable. If nothing qualifies, explain the limiting requirement and let the buyer edit it.

Build priority

Must work: prompt and confirmation; isolated seller proposals; server-enforced product and pricing rules; one useful challenge; optional single final-price revision; a clear comparison; source links and gaps; recheck/reservation; explicit approval; truthful simulated result.

After the core path works: one genuine non-tech evidence example, secondary profile checks, a small seller-policy view and a supported Visa test. Keep other categories in the model without claiming full live coverage.

Defer from the live path: accessory bundles, full seller administration, a second verification round and optional sponsor integrations. No real-money purchases or fulfillment are part of this prototype. These cuts are proposed changes to the previous plan.

Implementation, in one paragraph

Use the proposed React interface, Convex for stored state and server checks, Gemini for intake and seller assistance, and Apify for applicable public evidence. Keep totals, eligibility and purchase authority in server logic. Model and source access must be tested before the demo; source failures show a truthful gap. Use actual integrations only where they perform necessary work.

Trust, proof and the Visa story

Promises the prototype must enforce

- **Your maximum stays private.** Sellers receive a confirmed summary without the raw prompt or private ceiling. Shared targets require consent; perfect secrecy against inference is not promised.
- **Sellers control concessions.** Server rules reject unauthorized prices, unsupported stock and policy changes.
- **Evidence has a scope.** Cite applicable facts and show gaps. Product reviews do not certify a refurbished unit or seller.
- **You approve the purchase.** Recheck exact terms, require explicit approval and prevent duplicate execution. Changed terms need renewed approval.
- **The demo tells the truth.** Label fictional commerce, live/cached evidence and simulation. Collect no real card data in simulation.

Proof to bring to judging

Visa criterion	Show
Innovation, 25%	A buyer need becomes a suitable competing offer, with one correction or concession that matters.
User experience, 25%	Prompt, concise confirmation, useful comparison and exact-term approval within the pitch.
Technical execution, 20%	Actual scoped tool calls, a server-rejected invalid bid and persisted progress.
Commerce impact, 15%	Measured comparison time and simulated price/contribution outcomes with stated baselines.
Trust and responsibility, 15%	Private budget handling, applicable citations, honest gaps and buyer-controlled execution.

The Visa listing does not state an API mandate; confirm supplemental conditions. Enable a Visa test only after verifying its connection to the approved flow. Otherwise finish with “Demo order completed; no payment taken.” Tokenization is not payment.

Three measures, honestly reported

1. Time from confirmed need to valid comparison, separating cold and cached runs.
2. Constraint and policy correctness across a small disclosed request suite, including failures.
3. Initial-to-final delivered price and simulated seller contribution for the same candidate and terms.

Compare agents with list-price matching and automatic maximum discounts. Report useful corrections, contribution, cost and delay. Fictional transactions cannot prove conversion, loyalty or real profitability.

Three-minute demo

0:00-0:25: need and confirmed brief. **0:25-1:15:** offers and a useful correction. **1:15-2:10:** comparison, actual concession and cited tradeoff. **2:10-2:40:** recheck, approval and result. **2:40-3:00:** measurements and limits. Proposed presentation budget; runtime is unmeasured. Keep a labeled recording as fallback.

What remains uncertain

Will merchants participate and honor offers after real costs? Does the experience beat normal shopping, including waiting? Is evidence coverage reliable? A consented buyer/merchant pilot must test these assumptions.

TLDR: prioritize one complete shopping journey, useful agent work and honest simulation. Implementation, speed and market outcomes remain unverified.